LUIS PEDRO COELHO
Curriculum Vitæ
23 Aug 2026

Centre for Microbiome Research
Queensland University of Technology
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ORCID : 0000-0002-9280-7885
Erdös-Bacon Nr. : 7
Citizenship : U.K., Portugal, and Luxembourg

Education

2011 PhD in computational biology, Carnegie Mellon University
Dissertation topic: *Modeling the Space of Subcellular Location Patterns Using Images and Other Sources of Information*, advised by Prof. Robert F. Murphy.

2006 MSc in computer science, Instituto Superior Técnico (Technical University Lisbon)
Dissertation topic: *Bayesian Network Parameter Estimation Using Noisy Observations or Soft Evidence*, advised by Prof. Arlindo Oliveira.

2004 BSc in computer science, Instituto Superior Técnico (Technical University Lisbon)
Graduated top of my class.

Professional experience

2023–present Associate Professor (Queensland University of Technology)

2018–2023 Junior Principal Investigator (Fudan University)

2013–2018 Postdoctoral researcher at European Molecular Biology Laboratory (EMBL), Bork Lab

2012 Postdoctoral researcher at Instituto de Medicina Molecular (Lisbon), Mhlanga Lab

Scholarships & awards

2024 QUT Faculty of Health Researcher of the Year

2023 ARC Future Fellowship

2022–present Highly Cited Researcher (*Cross-field* category)

2021 Zhicheng Teaching Award (first prize)

2012 Siebel Scholar
Awarded annually for academic excellence and demonstrated leadership to 85 top students from the world's leading graduate schools

2009 Joint CMU-U. of Pittsburgh PhD. in Computational Biology Research Excellence Award

2008 Joint CMU-U. of Pittsburgh PhD. in Computational Biology Academic Excellence Award

2006 Fulbright Fellow

2005 Scholarship from Portuguese Science Foundation

2001 Instituto Superior Técnico (IST) Academic Excellence Award

Highlighted publications

2024 *C.D. Santos-Júnior*, M. D. Torres*, ..., C. de la Fuente-Nunez†*, **L. P. Coelho**†Discovery of antimicrobial peptides in the global microbiome with machine learning *in* CELL *[DOI]*

2022 **L. P. Coelho** *et al.* Towards the biogeography of prokaryotic genes *in* NATURE *[DOI]*

2022 *S. Pan, ...,* **L. P. Coelho** A deep siamese neural network improves metagenome-assembled genomes in microbiome datasets across different environments *in* NATURE COMMUNICATIONS *[DOI]*

2015 *S. Sunagawa**, **L. P. Coelho***, *S. Chaffron* et al.,* Structure and Function of the Global Ocean Microbiome *in* SCIENCE *[DOI]*

Grants awarded (selected)

2023 Surveillance for Emerging Antimicrobial Resistance through Characterization of the uncharted Environmental Resistome (SEARCHER)

2020 Establishing a Monitoring Baseline for Antibiotic Resistance in Key environments (EMBARK)

2019 Using deep learning to understand the microbiome, National Science Foundation of China

Service & outreach (selected)

2026 Chair of the National Conference Committee for the Australian Bioinformatics And Computational Biology Society (ABACBS)

2025 Member of the Executive Committee for the Australian Bioinformatics And Computational Biology Society (ABACBS)

2025–present Associate Editor for *Microbiome*

2021–2024 Member of the steering committee for mVIF (Microbiome Virtual International Forum)

2021–2023 Academic Editor for PLoS Computational Biology

2020–present Member of the ISCB (International Society for Computational Biology) Equity, Diversity and Inclusion (EDI) Committee

2016–2021 Associate Editor for the Journal of Open Research Software

2014–2017 Postdoc representative at EMBL
I helped organize the 2015 and 2017 EMBL Postdoc retreats.

2014–2015 Organized Software Carpentry Workshop at EMBL

2012–2013 Member of the Organization of the Lisbon Machine Learning School

2010 Local Committee for Portuguese-American Postgraduate Society National Forum

Teaching & mentoring (selected)

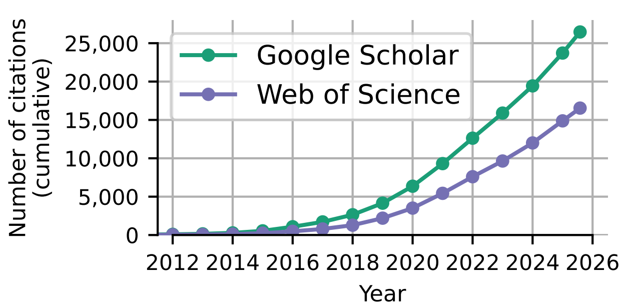
2019–2023 Co-taught *Scientific Communication* at Fudan University, a one-semester graduate course

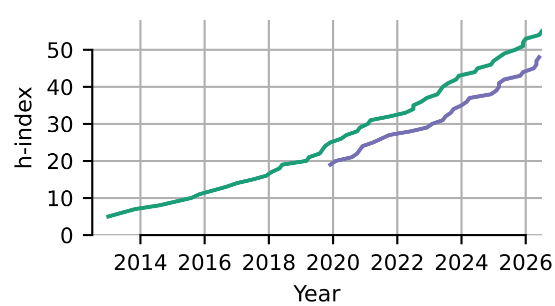
2012–present Certified Software Carpentry Instructor
I have given lectures on Software Carpentry in Germany, Denmark, Cyprus, Jordan, and Spain.

2009 Programming for Scientists
I designed and taught a semester-long course on computer programming for scientists at Carnegie Mellon University.

Bibliometric statistics

Total number of citations: 27,523 (Google Scholar); 16,869 (Web of Science)
h-index: 55 (Google Scholar); 48 (Web of Science)





Language skills

Native in **English** and **Portuguese**; fluent in **German** and **French**; intermediate **Luxembourgish**; basic **Auslan**.

Publications

Google scholar profile:
http://scholar.google.com/citations?user=qTYua0cAAAAJ

First or corresponding author

(includes co-first/co-corresponding author publications)

Preprints submitted for publication

1. Juan Salvador Inda-Díaz†, Faith Adegoke, Ulrike Löber, Víctor Hugo Jarquín-Díaz, Yiqian Duan, Johan Bengtsson-Palme, Svetlana Ugarcina Perovic, **Luis Pedro Coelho**†, *The elusive resistome: a global comparison reveals large discrepancies among detection pipelines*, in bioRxiv, 2026 [DOI]

2. Yiqian Duan, Anna Cusco, Yaozhong Zhang, Juan Salvador Inda-Diaz, Chengkai Zhu, Alexandre Areias Castro, Xinrun Yang, Jiabao Yu, Gaofei Jiang, Xing-Ming Zhao†, **Luis Pedro Coelho**†, *Long-read metagenomic sequencing reveals novel lineages and functional diversity in urban soil microbiomes*, in bioRxiv, 2026 [DOI]

3. Anna Cuscó†, Yiqian Duan, Fernando Gil, Alexei Chklovski, Nithya Kruthi, Shaojun Pan, Sofia Forslund, Susanne Lau, Ulrike Löber, Xing-Ming Zhao, **Luis Pedro Coelho**†, *Capturing global pet dog gut microbial diversity and hundreds of near-finished bacterial genomes by using long-read metagenomics in a Shanghai cohort*, in bioRxiv, 2025 [DOI]

4. Shaojun Pan, Ivan Tolstoganov, Kristoffer Sahlin, Marcel Martin, Xing-Ming Zhao, **Luis Pedro Coelho**, *AEMB: a computationally efficient abundance estimation method for metagenomic binning*, in bioRxiv, 2025 [DOI]

Peer-reviewed research publications

5. Svetlana Ugarcina Perovic, Vedanth Ramji, Hui Chong, Yiqian Duan, Finlay Maguire, **Luis Pedro Coelho**, *argNorm: normalization of antibiotic resistance gene annotations to the Antibiotic Resistance Ontology (ARO)*, in Bioinformatics, 2025 [DOI]

6. Yiqian Duan, Célio Dias Santos-Júnior, Thomas Sebastian Schmidt, Anthony Fullam, Breno L. S. de Almeida, Chengkai Zhu, Michael Kuhn, Xing-Ming Zhao†, Peer Bork, **Luis Pedro Coelho**†, *A catalog of small proteins from the global microbiome*, in Nature Communications, 2024 [DOI]

7. Célio Dias Santos-Júnior*, Marcelo D.T. Torres*, Yiqian Duan, Álvaro Rodríguez del Río, Thomas S.B. Schmidt, Hui Chong, Anthony Fullam, Michael Kuhn, Chengkai Zhu, Amy Houseman, Jelena Somborski, Anna Vines, Xing-Ming Zhao, Peer Bork, Jaime Huerta-Cepas, Cesar de la Fuente-Nunez†, **Luis Pedro Coelho**†, *Discovery of antimicrobial peptides in the global microbiome with machine learning*, in Cell, 2024 [DOI]

8. Marija Dmitrijeva, Janko Tackmann, João Frederico Matias Rodrigues, Jaime Huerta-Cepas, **Luis Pedro Coelho**†, Christian von Mering†, *A global survey of prokaryotic genomes reveals the eco-evolutionary pressures driving horizontal gene transfer*, in Nature Ecology & Evolution, 2024 [DOI]

9. Shaojun Pan, Xing-Ming Zhao†, **Luis Pedro Coelho**†, *SemiBin2: self-supervised contrastive learning leads to better MAGs for short- and long-read sequencing*, in Bioinformatics, 2023 [DOI]

10. Senying Lai, Shaojun Pan, **Luis Pedro Coelho**†, Wei-Hua Chen†, Xing-Ming Zhao†, *metaMIC: reference-free Misassembly Identification and Correction of de novo metagenomic assemblies* in Genome Biology 2022 [DOI]

11. Shaojun Pan, Chengkai Zhu, Xing-Ming Zhao†, and **Luis Pedro Coelho**† *A deep siamese neural network improves metagenome-assembled genomes in microbiome datasets across different environments* in Nature Communications 2022 [DOI]

12. **Luis Pedro Coelho**, Renato Alves, Álvaro Rodríguez del Río, Pernille Neve Myers, Carlos P. Cantalapiedra, Joaquín Giner-Lamia,Thomas Sebastian Schmidt, Daniel R. Mende, Askarbek Orakov, Ivica Letunic, Falk Hildebrand, Thea Van Rossum, Sofia K. Forslund, Supriya Khedkar, Oleksandr M. Maistrenko, Shaojun Pan, Longhao Jia, Pamela Ferretti, Shinichi Sunagawa, Xing-Ming Zhao, Henrik Bjørn Nielsen, Jaime Huerta-Cepas+, and Peer Bork+, *Towards the biogeography of prokaryotic genes* in NATURE 2022 [DOI]

13. Célio Dias Santos-Junior, Shaojun Pan, Xing-Ming Zhao, **Luis Pedro Coelho**, *MACREL: antimicrobial peptide screening in genomes and metagenomes* in PEERJ 2020 [DOI]

14. **Luis Pedro Coelho**, Renato Alves, Paulo Monteiro, Jaime Huerta-Cepas, Ana Teresa Freitas, Peer Bork *NG-meta-profiler: fast processing of metagenomes using NGLess, a domain-specific language* in MICROBIOME vol. 7, 84, 2019 [DOI]

15. **Luis Pedro Coelho**, Jens Kultima, Paul Costea, Coralie Fournier, Yuanlong Pan, Gail Czarnecki-Maulden, Matthew Hayward, Kristoffer Forslund, Patrick Descombes, Janet Jackson, Qinghong Li, and Peer Bork *Similarity of the dog and human gut microbiomes in gene content and response to diet* in MICROBIOME, vol 6:72, 2018 [DOI]

16. **Luis Pedro Coelho**, *Jug: Software for parallel reproducible computation in Python* in JOURNAL OF OPEN RESEARCH SOFTWARE, 2017 [DOI]

17. Sebastien Colin*, **Luis Pedro Coelho***, Shinichi Sunagawa, Chris Bowler, Eric Karsenti, Peer Bork, Rainer Pepperkok, Colomban de Vargas, *Quantitative 3D-imaging for cell biology and ecology of environmental microbial eukaryotes* in ELIFE, vol. 6:e26066, 2017 [DOI]

18. Shinichi Sunagawa*, **Luis Pedro Coelho***, Samuel Chaffron*, Jens Roat Kultima, Karine Labadie, Guillem Salazar, Bardya Djahanschiri, Georg Zeller, Daniel R. Mende, Adriana Alberti, Francisco M. Cornejo-Castillo, Paul I. Costea, Corinne Cruaud, Francesco d'Ovidio, Stefan Engelen, Isabel Ferrera, Josep M. Gasol, Lionel Guidi, Falk Hildebrand, Florian Kokoszka, Cyrille Lepoivre, Gipsi Lima-Mendez, Julie Poulain, Bonnie T. Poulos, Marta Royo-Llonch, Hugo Sarmento, Sara Vieira-Silva, Céline Dimier, Marc Picheral, Sarah Searson, Stefanie Kandels-Lewis, Tara Oceans coordinators, Chris Bowler, Colomban de Vargas, Gabriel Gorsky, Nigel Grimsley, Pascal Hingamp, Daniele Iudicone, Olivier Jaillon, Fabrice Not, Hiroyuki Ogata, Stephane Pesant, Sabrina Speich, Lars Stemmann, Matthew B. Sullivan, Jean Weissenbach, Patrick Wincker, Eric Karsenti, Jeroen Raes, Silvia G. Acinas, Peer Bork, *Structure and Function of the Global Ocean Microbiome* in SCIENCE 348 (6237), 1261359, 2015 [DOI]

19. **Luis Pedro Coelho**, Catarina Pato, Ana Friães, Ariane Neumann, Maren von Köckritz-Blickwede, Mário Ramirez, João André Carriço, *Automatic determination of NET (neutrophil extracellular traps) coverage in fluorescent microscopy images* in BIOINFORMATICS 31 (14): 2364–2370, 2015 [DOI]

20. **Luis Pedro Coelho**, Joshua D. Kangas, Armaghan Naik, Elvira Osuna-Highley, Estelle Glory-Afshar, Margaret Fuhrman, Ramanuja Simha, Peter B. Berget, Jonathan W. Jarvik, and Robert F. Murphy, *Determining the subcellular location of new proteins from microscope images using local features* in BIOINFORMATICS, 2013 [DOI]

21. **Luis Pedro Coelho** *Mahotas: Open source software for scriptable computer vision*, JOURNAL OF OPEN RESEARCH SOFTWARE, vol. 1, 2013 [DOI]

22. **Luis Pedro Coelho***, Tao Peng*, and Robert F. Murphy, *Quantifying the distribution of probes between subcellular locations using unsupervised pattern unmixing* in BIOINFORMATICS, vol. 26 (12), pp. i7–i12, 2010 [DOI]

23. **Luis Pedro Coelho**, Amr Ahmed, Andrew Arnold, Joshua Kangas, Abdul-Saboor Sheikh, Eric P. Xing, William W. Cohen, and Robert F. Murphy, *Structured Literature Image Finder: Extracting Information from Text and Images in Biomedical Literature* in LECTURE NOTES IN BIOINFORMATICS, vol. 6004, pp. 23–32, 2010 [DOI]

24. **Luis Pedro Coelho**, Aabid Shariff, and Robert F. Murphy; *Nuclear segmentation in microscope cell images: A hand-segmented dataset and comparison of algorithms* in PROCEEDINGS OF IEEE INTERNATIONAL SYMPOSIUM IN BIOMEDICAL IMAGING, 2009 [DOI]

25. **Luis Pedro Coelho** and Robert Murphy; *Identifying Subcellular Locations from Images of Unknown Resolution* in BIOINFORMATICS RESEARCH AND DEVELOPMENT, Communications in Computer and Information Science, vol. 13, pp. 235–242, 2008 [DOI]

26. **Luis Pedro Coelho** and Arlindo Oliveira; *Dotted Suffix Trees: A Structure for Approximate Text Indexing* in STRING PROCESSING AND INFORMATION RETRIEVAL, Lecture Notes in Computer Science, vol. 4209, pp. 329–336, 2006 [DOI]

Comments and review articles

27. **Luis Pedro Coelho**, *For long-term sustainable software in bioinformatics*, in PLOS COMPUTATIONAL BIOLOGY, 2024 [DOI]

28. **Luis Pedro Coelho**, Célio Dias Santos-Júnior, Cesar de la Fuente-Nunez, *Challenges in computational discovery of bioactive peptides in 'omics data*, in PROTEOMICS, 2024 [DOI]

29. **Luis Pedro Coelho** *Voices of the new generation: science in a state of benign confusion* in NATURE REVIEWS MOLECULAR CELL BIOLOGY 2020 [DOI]

30. **Luis Pedro Coelho**, Estelle Glory-Afshar, Joshua Kangas, Shannon Quinn, Aabid Shariff, and Robert F. Murphy; *Principles of Bioimage Informatics: Focus on machine learning of cell patterns* in LINKING LITERATURE, INFORMATION, AND KNOWLEDGE FOR BIOLOGY, Lecture Notes in Computer Science, vol. 6004, pp. 8–18, 2010 [DOI]

Co-authorships

Preprints submitted for publication

31. Lombard Fabien, Guidi Lionel, Manoela C. Brandão, Coelho Luis Pedro, Colin Sébastien, Dolan John Richard, Elineau Amanda, Josep M Gasol, Grondin Pierre Luc, Henry Nicolas, Federico M Ibarbalz, Jalabert Laëtitia, Loreau Michel, Martini Séverinne, Mériguet Zoé, Picheral Marc, Juan José Pierella Karlusich, Rainer Pepperkok, Romagnan Jean-Baptiste, Zinger Lucie, Stemmann Lars, Silvia G Acinas, Karp-Boss Lee, Boss Emmanuel, Matthew B. Sullivan, Colomban de Vargas, Bowler Chris, Karsenti Eric, Gorsky Gabriel, Tara Oceans Coordinators, *Ubiquity of inverted 'gelatinous' ecosystem pyramids in the global ocean*, in BIORXIV, 2024 [DOI]

Peer-reviewed research publications

32. Daniel R. Howard, Ridwan B. Rashid, Tufael Ahmed, Patricia Namubiru, Md. Sohel Rana, Tyler Wilkins, Joshua Morrow, Darren J. Creek, Rose Ann Franco, Mark C. Allenby, Declan L. Turner, Rhiannon B. Werder, Sam Manna, Catherine Satzke, Yenkai Lim, Jiarui Sun, Paul G. Dennis, Anushka Yadav, Andrew J. Kueh, Cheong Kwong Chung, Elizabeth Forbes-Blom, Mario Noti, Jonathan O'Regan, **Luis Pedro Coelho**, Chrysothemis C. Brown, Mark Morrison, Simon Phipps, *Milk osteopontin alters the infant microbiome to drive DC hematopoiesis and disease tolerance*, in CELL, 2026 [DOI]

33. Iakovos C. Iakovides, Sotirios Vasileiadis, Anastasis Christou, Popi Karaolia, Theoni Mina, Jaqueline Rocha, Yiqian Duan, Vasiliki G. Beretsou, Nicolas Gallois, Frédérique Changey, Costas Michael, **Luis Pedro Coelho**, Celia M. Manaia, Christophe Merlin, Despo Fatta-Kassinos, *Storage and soil depth, in addition to wastewater treatment, govern microbiota, and mobile genetic element and antibiotic resistance markers during reclaimed water irrigation*, in WATER RESEARCH, 2026 [DOI]

34. Vishnu Prasoodanan PK, Oleksandr M. Maistrenko, Anthony Fullam, Daniel R. Mende, Ece Kartal, **Luis Pedro Coelho**, Anja Spang, Peer Bork, Thomas S. B. Schmidt, *Unbinned contigs expand known diversity in the global microbiome*, in NATURE MICROBIOLOGY, 2026 [DOI]

35. John Paul Makumbi, Samuel K. Leareng, Oliver K. Bezuidt, **Luis Pedro Coelho**, Thulani P. Makhalanyane, *Persistence of high-risk antimicrobial resistance genes in extracellular DNA along an urban wastewater-river continuum*, in CELL REPORTS, 2026 [DOI]

36. Kanta Chechi, Rima Chakaroun, Antonis Myridakis, Sofia K. Forslund-Startceva, Sebastien Fromentin, Trine Nielsen, Judith Aron-Wisneswky, Eugeni Belda, Edi Prifti, Pierre Bel Lassen, Gwen Falony, Sara Vieira-Silva, Julien Chilloux, Kazuhiro Sonomura, Lesley Hoyles, Laura Martinez-Gili, Francesco Pallotti, Petros Andrikopoulos, Francesc Puig-Castellví, Romina Pacheco Tapia, Inés Castro-Dionicio, Hugo Roume, Nicolas Pons, Emmanuelle Le Chatelier, Benoit Quinquis, Nathalie Galleron, Magali Berland, Michael T. Olanipekun, Manyi Jia, Angelos Manolias, Bridget Holmes, Solia Adriouch, Matthias Blüher, **Luis Pedro Coelho**, Kévin Da Silva, Pilar Galan, Boyang Ji, Ana Luisa Neves, Christine Rouault, Joe-Elie Salem, Valentina Tremaroli, Tue H. Hansen, Nadja B. Søndertoft, Christian Lewinter, Helle K. Pedersen, The MetaCardis Consortium, Rohia Alili, Chloe Amouyal, Karen Assmann, Ehm Astrid Andersson Galijatovic, Fabrizio Andreelli, Olivier Barthelemy, Jean-Philippe Bastard, Jean-Paul Batisse, Randa Bittar, Hervé Blottière, Frederic Bosquet, Rachid Boubrit, Olivier Bourron, Mickael Camus, Dominique Cassuto, Cécile Ciangura, Jean-Philippe Collet, Maria-Carlota Dao, Morad Djebbar, Angélique Doré, Line Engelbrechtsen, Soraya Fellahi, Leopold Fezeu, Philippe Giral, Agnes Hartemann, Bolette Hartmann, Gerard Helft, Jens Juul Holst, Marlene Hornbak, Andrea Rodriguez-Martinez, Jean-Sebastien Hulot, Richard Isnard, Sophie Jaqueminet, Niklas Rye Jørgensen, Hanna Julienne, Johanne Justesen, Judith Kammer, Nikolaj Karup, Mathieu Kerneis, Jean Khemis, Michael Kuhn, Véronique Lejard, Florence Levenez, Lea Lucas-Martini, Robin Massey, Nicolas Maziers, Jonathan Medina-Stamminger, Gilles Montalescot, Sandrine Moutel, Laetitia Pasero Le Pavin, Christine Poitou-Bernert, Francoise Pousset, Laurence Pouzoulet, Lucas Moitinho-Silva, Johanne Silvain, Mathilde Svendstrup, Timothy Swartz, Thierry Vanduyvenboden, Camille Vatier, Stefanie Walther, Eric Verger, Aurélie Lampure, Peter D. Mark, Jens P. Goetze, Lars Køber, Henrik Vestergaard, Torben Hansen, Jean-Daniel Zucker, Taka-Aki Sato, Serge Hercberg, Fredrik Bäckhed, Ivica Letunic, Jean-Michel Oppert, Jens Nielsen, Jeroen Raes, Ioanna Tzoulaki, Abbas Dehghan, Verena Zuber, Emmanuelle Bouzigon, Mark Lathrop, Parminder Raina, Philippe Froguel, Fumihiko Matsuda, Florence Demenais, Dominique Gauguier, Michael Stumvoll, Peer Bork, Oluf Pedersen, S. Dusko Ehrlich, Karine Clément, Marc-Emmanuel Dumas, *A gut microbiome-kidney-heart axis predictive of future cardiovascular diseases*, in Nature Communications, 2026 [DOI]

37. Anthony Fullam, Ivica Letunic, Oleksandr M Maistrenko, Alexandre Areias Castro, **Luis Pedro Coelho**, Anastasiia Grekova, Christian Schudoma, Supriya Khedkar, Mahdi Robbani, Michael Kuhn, Thomas S B Schmidt, Peer Bork, Daniel R Mende, *proGenomes4: providing 2 million accurately and consistently annotated high-quality prokaryotic genomes*, in Nucleic Acids Research, 2025 [DOI]

38. Álvaro Rodríguez del Río, Joaquín Giner-Lamia, Carlos P. Cantalapiedra, Jorge Botas, Ziqi Deng, Ana Hernández-Plaza, Martí Munar-Palmer, Saray Santamaría-Hernando, José J. Rodríguez-Herva, Hans-Joachim Ruscheweyh, Lucas Paoli, Thomas S. B. Schmidt, Shinichi Sunagawa, Peer Bork, Emilia López-Solanilla, **Luis Pedro Coelho**, Jaime Huerta-Cepas *Functional and evolutionary significance of unknown genes from uncultivated taxa* in Nature, 2023 [DOI]

39. Thomas S B Schmidt, Anthony Fullam, Pamela Ferretti, Askarbek Orakov, Oleksandr M Maistrenko, Hans-Joachim Ruscheweyh, Ivica Letunic, Yiqian Duan, Thea Van Rossum, Shinichi Sunagawa, Daniel R Mende, Robert D Finn, Michael Kuhn, **Luis Pedro Coelho**, Peer Bork, *SPIRE: a Searchable, Planetary-scale mIcrobiome REsource*, in Nucleic Acids Research, 2023 [DOI]

40. Rémi Gschwind, Svetlana Ugarcina Perovic, Marie Petitjean, Julie Lao, **Luis Pedro Coelho**, Etienne Ruppé *ResFinderFG v2.0: a database of antibiotic resistance genes obtained by functional metagenomics* in Nucleic Acid Research 2023 [DOI]

41. Hui Chong, Qingyang Yu, Yuguo Zha, Guangzhou Xiong, Nan Wang, Xinhe Huang, Shijuan Huang, Chuqing Sun, Sicheng Wu, Wei-Hua Chen, **Luis Pedro Coelho**, Kang Ning *EXPERT: Transfer Learning-enabled context-aware microbial source tracking* in Briefings in Bioinformatics 2022 [DOI]

42. Thomas SB Schmidt, Simone S Li, Oleksandr M Maistrenko, Wasiu Akkani, **Luis Pedro Coelho**, Sibasish Dolai, Anthony Fullam, Anna Glazek, Rajna Hercog, Hilde Herrema, Ferris Jung, Stefanie Kandels, Askarbek Orakov, Thea Van Rossum, Vladimir Benes, Thomas J Borody, Willem M de Vos, Cyriel Y Ponsioen, Max Nieuwdorp, Peer Bork *Drivers and Determinants of Strain Dynamics Following Faecal Microbiota Transplantation* in Nature Medicine 2022 [DOI]

43. Sebastien Fromentin, Sofia K. Forslund, Kanta Chechi, Judith Aron-Wisnewsky, Rima Chakaroun, Trine Nielsen, Valentina Tremaroli, Boyang Ji, Edi Prifti, Antonis Myridakis, Julien Chilloux, Petros Andrikopoulos, Yong Fan, Michael T. Olanipekun, Renato Alves, Solia Adiouch, Noam Bar, Yeela Talmor-Barkan, Eugeni Belda, Robert Caesar, **Luis Pedro Coelho**, Gwen Falony, Soraya Fellahi, Pilar Galan, Nathalie Galleron, Gerard Helft, Lesley Hoyles, Richard Isnard, Emmanuelle Le Chatelier, Hanna Julienne, Lisa Olsson, Helle Krogh Pedersen, Nicolas Pons, Benoit Quinquis, Christine Rouault, Hugo Roume, Joe-Elie Salem, Thomas S. B. Schmidt, Sara Vieira-Silva, Peishun Li, Maria Zimmermann-Kogadeeva, Christian Lewinter, Nadja B. Søndertoft, Tue H. Hansen, Dominique Gauguier, Jens Peter Gøtze, Lars Køber, Ran Kornowski, Henrik Vestergaard, Torben Hansen, Jean-Daniel Zucker, Serge Hercberg, Ivica Letunic, Fredrik Bäckhed, Jean-Michel Oppert, Jens Nielsen, Jeroen Raes, Peer Bork, Michael Stumvoll, Eran Segal, Karine Clément†, Marc-Emmanuel Dumas†, S. Dusko Ehrlich†, and Oluf Pedersen† *Microbiome and metabolome features of the cardiometabolic disease spectrum* in NATURE MEDICINE 2022 [DOI]

44. Supriya Khedkar, Georgy Smyshlyaev, Ivica Letunic, Oleksandr M Maistrenko, **Luis Pedro Coelho**, Askarbek Orakov, Sofia K Forslund, Falk Hildebrand, Mechthild Luetge, Thomas S B Schmidt, Orsolya Barabas, and Peer Bork *Landscape of mobile genetic elements and their antibiotic resistance cargo in prokaryotic genomes* in NUCLEIC ACIDS RESEARCH 2022 [DOI]

45. Eugeni Belda*, Lise Voland*, Valentina Tremaroli*, Gwen Falony*, Solia Adriouch*, Karen E Assmann*, Edi Prifti, Judith Aron-Wisnewsky, Jean Debédat, Tiphaine Le Roy, Trine Nielsen, Chloé Amouyal, Sébastien André, Fabrizio Andreelli, Matthias Blüher, Rima Chakaroun, Julien Chilloux, **Luis Pedro Coelho**, Maria Carlota Dao, Promi Das, Soraya Fellahi, Sofia Forslund, Nathalie Galleron, Tue H Hansen, Bridget Holmes, Boyang Ji, Helle Krogh Pedersen, Phuong Le, Emmanuelle Le Chatelier, Christian Lewinter, Louise Mannerås-Holm, Florian Marquet, Antonis Myridakis, Veronique Pelloux, Nicolas Pons, Benoit Quinquis, Christine Rouault, Hugo Roume, Joe-Elie Salem, Nataliya Sokolovska, Nadja B Søndertoft, Sothea Touch, Sara Vieira-Silva, The MetaCardis Consortium, Pilar Galan, Jens Holst, Jens Peter Gøtze, Lars Køber, Henrik Vestergaard, Torben Hansen, Serge Hercberg, Jean-Michel Oppert, Jens Nielsen, Ivica Letunic, Marc-Emmanuel Dumas, Michael Stumvoll, Oluf Borbye Pedersen, Peer Bork, Stanislav Dusko Ehrlich, Jean-Daniel Zucker, Fredrik Bäckhed, Jeroen Raes, and Karine Clément *Impairment of gut microbial biotin metabolism and host biotin status in severe obesity: effect of biotin and prebiotic supplementation on improved metabolism* in GUT 2022 [DOI]

46. Boas C.L. van der Putten*, C. I. Mendes*, Brooke M. Talbot, Jolinda de Korne-Elenbaas, Rafael Mamede, Pedro Vila-Cerqueira, **Luis Pedro Coelho**, Christopher A. Gulvik, Lee S. Katz, and The ASM NGS 2020 Hackathon participants *Software testing in microbial bioinformatics: a call to action* in MICROBIAL GENOMICS 2022 [DOI]

47. Sofia K. Forslund*, Rima Chakaroun*, Maria Zimmermann-Kogadeeva*, Lajos Markó*, Judith Aron-Wisnewsky*, Trine Nielsen*, Lucas Moitinho-Silva, Thomas S. B. Schmidt, Gwen Falony, Sara Vieira-Silva, Solia Adriouch, Renato J. Alves, Karen Assmann, Jean-Philippe Bastard, Till Birkner, Robert Caesar, Julien Chilloux, **Luis Pedro Coelho**, Leopold Fezeu, Nathalie Galleron, Gerard Helft, Richard Isnard, Boyang Ji, Michael Kuhn, Emmanuelle Le Chatelier, Antonis Myridakis, Lisa Olsson, Nicolas Pons, Edi Prifti, Benoit Quinquis, Hugo Roume, Joe-Elie Salem, Nataliya Sokolovska, Valentina Tremaroli, Mireia Valles-Colomer, Christian Lewinter, Nadja B Søndertoft, Helle Krogh Pedersen, Tue H Hansen, *The MetaCardis Consortium*, Jens Peter Gøtze, Lars Køber, Henrik Vestergaard, Torben Hansen, Jean-Daniel Zucker7,20,21, Serge Hercberg, Jean-Michel Oppert, Ivica Letunic, Jens Nielsen, Fredrik Bäckhed, S. Dusko Ehrlich, Marc-Emmanuel Dumas, Jeroen Raes, Oluf Pedersen, Karine Clément†, Michael Stumvoll†, Peer Bork† *Combinatorial, additive and dose-dependent drug- microbiome associations* in NATURE 2021 [DOI]

48. Askarbek Orakov*, Anthony Fullam*, **Luis Pedro Coelho**, Supriya Khedkar, Damian Szklarczyk, Daniel R Mende, Thomas SB Schmidt† and Peer Bork† *GUNC: Detection of Chimerism and Contamination in Prokaryotic Genomes* in GENOME BIOLOGY 2021 [DOI]

49. Mohammad Bahram, Tarquin Netherway, Clémence Frioux, Pamela Ferretti, **Luis Pedro Coelho**, Stefan Geisen, Peer Bork, and Falk Hildebrand *Metagenomic assessment of the global diversity and distribution of bacteria and fungi* in ENVIRONMENTAL MICROBIOLOGY 2020 [DOI]

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1. **Luis Pedro Coelho**, Willi Richert; *Building Machine Learning Systems with Python*, Packt Publishing, 2013 (first ed.); 2015 (second ed.); 2018 (third ed.)

Talks

Category	Total	Since 2022
Keynotes & invited conference talks	30	18
Institutional seminars	36	24
Contributed conference talks	14	7
Lectures, webinars & outreach	19	7
Total	**99**	**56**

Talks delivered in 27 countries (22 since 2022), including 3 conference keynotes. Additionally: session convenor and panellist (2 panels) at ISME20 (Auckland, New Zealand), 2026.

Selected Recent Talks

1. *AI for understanding small proteins and peptides from the global microbiome* at the 20th International Symposium on Microbial Ecology (ISME20), Auckland (New Zealand), August 2026

2. *Big data and small genes. Studying small proteins from the global microbiome* at the annual meeting of the Sociedade Brasileira de Bioquímica e Biologia Molecular (SBBq), Águas de Lindoia (Brazil), May 2026

3. *Big data and small genes. Studying small proteins from the global microbiome*, invited keynote at DTMBIO, Muju (South Korea), December 2025

4. *The global microbiome as a source of antimicrobial peptides* at the SMBE Satellite Meeting on the Evolutionary Biochemistry of Insect Antimicrobial Peptides, Houston (USA), October 2025

5. *Small proteins of the human microbiome and beyond* at the EMBL Human Microbiome Symposium, Heidelberg (Germany), September 2025

6. *Decoding the microproteins of the global microbiome with machine learning and gene synteny* at the Gordon Research Conference on Decoding Microproteins Across Evolution and Disease, Barcelona (Spain), August 2025

7. *Long-read metagenome-assembled genomes can be higher quality than reference genomes: the case of the Shanghai pet dog microbiome catalog* at ISMB/ECCB, Liverpool (United Kingdom), July 2025 (contributed talk)

8. *Big Data & Small Genes in the Global Microbiome*, seminars at Imperial College London, King's College London, and the University of Warwick (United Kingdom), July 2025

9. *How we develop sustainable software tools in an academic setting* at the Symposium on Bioinformatics Excellence and Innovation (ABACBS 2024), Sydney (Australia), November 2024

10. *SemiBin2: self-supervised learning leads to better MAGs for short- and long-reads* at ISMB/ECCB, Lyon (France), July 2023 (proceedings paper, MICROBIOME track); also presented in the Highlights track of InCoB, Brisbane (Australia), November 2023

11. *Analysing the microbiome at a global scale* at the Novo Nordisk Foundation Prize Symposium, Copenhagen (Denmark), June 2022

12. *Microproteins of the global microbiome* at the Keystone Symposium on Micropeptides: Biogenesis and Function, Salt Lake City (USA), April 2022

Selected Earlier Talks

1. *Microbes and antimicrobes at the global scale* at Institut Pasteur Shanghai (China), December 2020

2. *The use of animal models for assessing antimicrobial impact on the gut microbiome* at the International Conference on One Health Antimicrobial Resistance (ICOHAR), Utrecht (Netherlands), April 2019

3. *Metagenomics-based investigations of microbial communities* at the Symposium on Computational Biology, Fudan University, Shanghai (China), August 2018

4. *Looking at the oceans with computer vision in Python*, invited keynote at PyCon Sette, Florence (Italy), April 2016

5. *Environmental High-content Fluorescence Microscopy (e-HCFM) of Tara Oceans Samples Provides a View of Global Ocean Protist Biodiversity* at the Ocean Sciences Meeting, New Orleans (USA), February 2016 (contributed talk)

6. *Using theatre to fight HIV/AIDS in Mozambique* (with Rita Reis) at the National Conference of the Association for Theatre in Higher Education, Chicago (USA), 2011 (contributed talk)

7. *Determining resolvable subcellular location categories as a function of image resolution* at the 24th ISAC Congress, Budapest (Hungary), September 2008 (contributed talk)